

# BOM Manual Downloader

## User & Developer Documentation

Automatically fetches datasheets for every part in your Bill of Materials and merges them into one clean, ordered Manufacturer's Record Book PDF.

*Version: July 2026 | Platform: Windows 10/11 | License: MIT*

## 1. Overview

BOM Manual Downloader is a Windows desktop application that reads a Bill of Materials (PDF, Excel, Word, Markdown or text), extracts every manufacturer and part number, searches manufacturer websites and search engines for the matching datasheet or manual, downloads and verifies each PDF, and merges everything into a single professional “Manufacturer's Record Book” with a clickable table of contents, chapter divider pages, and page numbering.

The app runs fully offline-installable as a portable folder — no Python required on the target machine. A local “catalog” folder of already-collected datasheets can be attached so known parts are resolved instantly without any web search.

## 2. Key Features

- Reads PDF, Excel (XLSX/XLS), Word (DOCX), Markdown and TXT BOMs with automatic column detection.
- Catalog-first lookup: exact part hits in your own datasheet library skip the web entirely.
- Concurrent downloads (5 parallel workers by default) with live per-part status cards.
- Every downloaded PDF is content-verified to be a genuine datasheet/manual for that part.
- Duplicate part detection: repeated BOM lines are kept, highlighted, and can be kept or discarded; kept duplicates reuse the first occurrence's document without a second search.
- Per-part controls during a run: skip, re-queue, search again, upload your own PDF, auto-pick.
- Manufacturer's Record Book output: table of contents with clickable “Go to Datasheet” links, chapter breaker pages, Page X of Y footers.
- Selective merge: check exactly which found manuals go into the final PDF.
- Standalone Merge PDFs tool for combining arbitrary PDFs with drag-and-drop ordering.
- Works out of the box with free search (DuckDuckGo + direct manufacturer probing); optional API keys raise hit rates.

## 3. Getting Started

### 3.1 Portable app (recommended)

1. Copy the BomDownloader folder (or unzip BomDownloader-Portable.zip) anywhere on the PC.
2. Double-click BomDownloader.exe.
3. No installation and no Python needed. Settings are stored per-user under %APPDATA%\BomDownloader.

Target machine requirements: Windows 10/11 with the Microsoft WebView2 runtime (preinstalled on Windows 11 and on any PC with Microsoft Edge).

### 3.2 Running from source

1. Install Python 3.10 or newer.
2. `pip install -r requirements.txt`
3. `python main.py`

## 4. Using the App — the 4-Step Workflow

### Step 1 — Input

Drop a BOM document (PDF / XLSX / XLS / DOCX / Markdown / TXT) on the drop zone, or switch to “Parts File” mode to load a simple structured parts list. The parser detects manufacturer / part number / description columns automatically.

### Step 2 — Parts review

Every extracted part is shown in an editable table in original BOM order. You can edit any cell, reorder rows, delete rows, and add parts manually.

#### **Duplicate parts:**

- Rows whose manufacturer + part number repeat an earlier row are tinted amber and badged “DUP of #NNN”, pointing at the first occurrence.
- A banner shows the duplicate count with a one-click “Discard all duplicates” button; individual duplicates can be removed with the row's delete button.
- Duplicates you keep are never searched twice — when the original part's document is found, the duplicate receives its own copy so the merged PDF repeats the document at the duplicate's exact BOM position.
- Badges update live while you edit — changing a part number in or out of duplication immediately updates the highlighting.

The collapsible “Download settings” panel configures output folder, catalog folder(s), max PDFs per part, parallel workers, custom logo, and skip-merge.

### Step 3 — Download

A live grid shows one card per part: Pending → Searching → Found / Not found. While running you can cancel the whole job, skip single parts, or re-queue parts. Duplicates show their badge and flip to “Found (duplicate)” the moment their original resolves. An activity log streams every search and download step. Parts that found nothing in the parallel pass are automatically retried one-at-a-time with search engines forced.

### Step 4 — Results

A summary table lists every part with Found / Not found / Skipped status. For each found part you can open the PDF, search again (pick a different document, upload your own, or auto-pick).

Check the manuals you want and press Merge to produce BOM\_Manuals\_Selected.pdf; the full BOM\_Manuals\_Merged.pdf record book is created automatically when the run completes. Cancelling mid-run still gives usable partial results.

## 5. The Manufacturer's Record Book

The merged output reproduces INTECH's record book format:

- Cover: black TABLE OF CONTENTS bar and a ruled grid — Chapter #, Item Description, Manufacturer, Part No., and a clickable blue “Go to Datasheet” link per row.
- A breaker page before every manual showing the chapter number, manufacturer and part.
- “Page X of Y” footer on all generated pages; TOC links jump to the right chapter.
- Parts whose document was not found get a “Document Not Found” placeholder page, so the book always matches the BOM row-for-row.

## 6. Using a Datasheet Catalog

Point “My catalog” at one or more sources (separate several with ;): a local folder, an OneDrive/SharePoint “Anyone with the link” share, or a public Google Drive folder. The app indexes every PDF and, before any web search, checks for a confident hit — an exact part number in a filename, inside a catalog PDF, or an explicit pin in catalog\_map.txt (one “part = filename fragment” per line). Confident hits skip the web entirely; fuzzy matches are only used as a clearly-flagged last resort.

## 7. Settings Reference

| Setting           | Default               | Description                                              |
|-------------------|-----------------------|----------------------------------------------------------|
| Output folder     | Documents\BOM Manuals | Where downloads and merged PDFs are saved                |
| My catalog        | (empty)               | Local folder / OneDrive / Google Drive datasheet library |
| Max PDFs per part | 1                     | How many documents to keep per part (1–5)                |
| Parallel workers  | 5                     | Concurrent download threads (1–20)                       |
| Custom logo       | (default)             | Replaces the logo on generated pages                     |
| Skip merging      | Off                   | Download files only, no record book                      |

## 8. Optional Search Backends

The app works with no configuration using DuckDuckGo plus direct manufacturer-site probing. For higher hit rates on large BOMs, API keys can be added at the top of bom\_downloader.py: Nexar, Google CSE, Bing, Mouser, Farnell, Serper, SerpApi, Tavily, Exa, Brave, DigiKey. A circuit breaker disables any backend after repeated failures so one bad key never slows a run.

## 9. Building the Portable App

1. Run build.bat from the project root (installs dependencies, runs PyInstaller with BomDownloader.spec, and zips the result).
2. Outputs: dist\BomDownloader\ (portable folder) and dist\BomDownloader-Portable.zip.

**Always build via BomDownloader.spec.**

A bare “pyinstaller main.py” does NOT bundle the ui\ folder or the SSL runtime hook — the resulting exe exits silently on launch. The spec also disables UPX (antivirus false-positives) and includes the WebView2 and python-docx hidden imports.

## 10. Troubleshooting

| Symptom                                   | Cause / Fix                                                                   |
|-------------------------------------------|-------------------------------------------------------------------------------|
| Exe exits silently on launch              | Built without the spec file — rebuild with build.bat / BomDownloader.spec     |
| Blank window on another PC                | WebView2 runtime missing — install Microsoft Edge WebView2                    |
| All parts stuck on Searching (old builds) | Fixed July 2026 — logging crash in frozen builds; update to the current build |
| Zero results everywhere                   | Check internet/proxy; see the activity log for per-backend errors             |
| A wrong document was picked               | Results page → Search again → choose another candidate or upload your own PDF |
| Antivirus quarantines the exe             | Use the current build (UPX disabled); whitelist the folder if needed          |

## 11. Project Structure

| File                           | Responsibility                                                                 |
|--------------------------------|--------------------------------------------------------------------------------|
| main.py                        | Entry point — boots the pywebview window                                       |
| api.py                         | Python ↔ JavaScript bridge; job orchestration, progress, duplicates resolution |
| bom_downloader.py              | Core engine: BOM parsing, search cascade, download, verification, record book  |
| ui/index.html                  | Complete frontend (HTML + CSS + JS, single file)                               |
| BomDownloader.spec / build.bat | PyInstaller build recipe / one-click build                                     |
| rthook_ssl.py                  | Frozen-build SSL certificate runtime hook                                      |
| tests/                         | Plain-assert regression test scripts                                           |